Mathematics Method for Physics—Lecture 9

The Inverse Laplace Transform and Its Applications

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Conventions: $p = \sigma + \mathrm{i}\omega$; $u(t)$ is the unit step function; $\mathrm{e}, \mathrm{i}, \mathrm{d}$ *upright; formulas are numbered (5.1)–(5.5).*

5 Inverse Laplace Transform and Applications

The core of integral-transform theory can be summarized in four items:

1. **Definition and theorems**: the Fourier transform (on $(-\infty, +\infty)$) and the Laplace transform (on $[0, +\infty)$), together with their existence theorems;
2. **Common transform pairs**: elementary functions;
3. **Properties**: linearity, similarity, delay (time shift), frequency shift, differentiation, integration, periodicity — about seven or eight basic properties (here too "seven or eight" is approximate; the definitive list is the one collected in Chapter 4);
4. **Applications**: solving differential and integral equations, and evaluating certain generalized integrals.

5.1 The Inverse Laplace Transform

5.1.1 The complex inversion integral formula

Everything rests on one observation: the Laplace transform of $f(t)$ is exactly the Fourier transform of $f(t)u(t)\,\mathrm{e}^{-\sigma t}$, where σ is a real constant. Indeed, $u(t)$ pulls the lower limit from $-\infty$ to 0,

$$\mathcal{F}\left[f(t)u(t)\,\mathrm{e}^{-\sigma t}\right] = \int_{-\infty}^{+\infty} f(t)u(t)\,\mathrm{e}^{-\sigma t}\,\mathrm{e}^{-\mathrm{i}\omega t}\,\mathrm{d}t = \int_{0}^{+\infty} f(t)\,\mathrm{e}^{-(\sigma+\mathrm{i}\omega)t}\,\mathrm{d}t = F(p), \qquad p = \sigma + \mathrm{i}\omega.$$

Applying the Fourier integral formula to $f(t)u(t)\,\mathrm{e}^{-\sigma t}$ gives

$$f(t)u(t)\,\mathrm{e}^{-\sigma t} = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\sigma + \mathrm{i}\omega)\,\mathrm{e}^{\mathrm{i}\omega t}\,\mathrm{d}\omega.$$

Multiply by $\mathrm{e}^{\sigma t}$ and use $u(t) = 1$ for $t > 0$; then substitute $p = \sigma + \mathrm{i}\omega$, so that $\mathrm{d}p = \mathrm{i}\,\mathrm{d}\omega$ and $\omega : -\infty \to +\infty$ corresponds to $p : \sigma - \mathrm{i}\infty \to \sigma + \mathrm{i}\infty$. This yields

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\sigma + \mathrm{i}\omega)\,\mathrm{e}^{(\sigma+\mathrm{i}\omega)t}\,\mathrm{d}\omega = \frac{1}{2\pi\mathrm{i}}\int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} F(p)\,\mathrm{e}^{pt}\,\mathrm{d}p \qquad (t > 0). \tag{5.1}$$

The factor in front differs between the two forms ($1/2\pi$ versus $1/(2\pi\mathrm{i})$); the difference is precisely $\mathrm{d}p = \mathrm{i}\,\mathrm{d}\omega$. The integral on the right is called the **Bromwich integral**; its path is any line $\mathrm{Re}\,p = \sigma$ parallel to the imaginary axis, lying in the region of convergence.

Theorem 5.1 (Complex inversion integral)**.** *Let $f(t)$ be piecewise smooth on $t \geq 0$ and of exponential order, $|f(t)| \leq M\mathrm{e}^{\sigma_c t}$, and let $\mathcal{L}[f(t)] = F(p)$. Take $\sigma > \sigma_c$. Then for $t > 0$*

$$f(t) = \frac{1}{2\pi\mathrm{i}} \int_{\sigma-\mathrm{i}\infty}^{\sigma+\mathrm{i}\infty} F(p)\,\mathrm{e}^{pt}\,\mathrm{d}p, \tag{5.2}$$

where at a discontinuity of f the left-hand side is to be replaced by the mean of the one-sided limits, $\frac{f(t+0)+f(t-0)}{2}$.

The condition $\sigma > \sigma_c$ makes $f(t)u(t)\,\mathrm{e}^{-\sigma t}$ absolutely integrable on $(-\infty, +\infty)$, which is what the Fourier integral formula requires; it also puts the path inside the region of convergence $\operatorname{Re} p > \sigma_0$ (σ_0 is the abscissa of convergence, $\sigma_0 \leq \sigma_c$). The image function must in addition satisfy $\lim_{p\to\infty} F(p) = 0$ — the necessary condition for a Laplace transform (Chapter 4, remark (1)).

5.1.2 Inversion by the residue theorem

The complex integral (5.2) is rarely evaluated directly; instead the path is closed to the left and the integral becomes a sum of residues.

Close the path with the left semicircle C_ρ of radius ρ centred at $\sigma + \mathrm{i}0$, which together with the vertical segment bounds a left half-disc. On C_ρ write $p = \sigma + \rho\,\mathrm{e}^{\mathrm{i}\theta}$ with $\pi/2 < \theta < 3\pi/2$; then

$$\left|\mathrm{e}^{pt}\right| = \mathrm{e}^{\sigma t}\mathrm{e}^{\rho t\cos\theta}, \qquad \cos\theta < 0.$$

The exponential factor decays along the arc, and the integral over the arc is bounded by $\pi\mathrm{e}^{\sigma t}/t$ (the mechanism behind Jordan's lemma). Hence, provided $F(p)$ tends to zero uniformly on the arc — guaranteed by $\lim_{p\to\infty} F(p) = 0$ — the arc integral tends to zero as $\rho \to +\infty$. For $t < 0$ the exponential factor forces the path to be closed to the right, where there are no singularities, so the integral vanishes; this is why (5.2) is stated for $t > 0$ only.

Theorem 5.2 (Inversion by residues)**.** *Let $F(p)$ have finitely many singular points $p_1, \ldots, p_n$, all to the left of the line $\operatorname{Re} p = \sigma$, and let $\lim_{p\to\infty} F(p) = 0$. Then for $t > 0$*

$$f(t) = \mathcal{L}^{-1}[F(p)] = \sum_{k=1}^{n} \operatorname{Res}\left[F(p)\,\mathrm{e}^{pt},\, p_k\right]. \tag{5.3}$$

Proof. Combine the path of (5.2) with the left semicircle into a counterclockwise closed contour containing every singularity of $F(p)\mathrm{e}^{pt}$. By the residue theorem the contour integral equals $2\pi\mathrm{i}\sum_k \operatorname{Res}$; letting $\rho \to +\infty$, the arc integral vanishes and the vertical segment tends to the path of (5.2), which gives (5.3). ∎

For a rational function $F(p) = \dfrac{A(p)}{B(p)}$ with $\deg B > \deg A$ all singularities are poles and $F(\infty) = 0$ holds automatically, so this is the case met in practice. If every p_k is a simple pole,

$$f(t) = \sum_{k=1}^{n} \frac{A(p_k)}{B'(p_k)}\,\mathrm{e}^{p_k t}; \tag{5.4}$$

if p_k is a pole of order m_k,

$$f(t) = \sum_{k=1}^{n} \frac{1}{(m_k-1)!} \lim_{p\to p_k} \frac{\mathrm{d}^{m_k-1}}{\mathrm{d}p^{m_k-1}} \left[(p-p_k)^{m_k} \frac{A(p)}{B(p)}\,\mathrm{e}^{pt}\right]. \tag{5.5}$$

Example 5.1. Find $\mathcal{L}^{-1}\left[\dfrac{1}{p^2+1}\right]$.

Solution. The poles $p = \pm\mathrm{i}$ are simple; by (5.4),

$$\mathcal{L}^{-1}\left[\frac{1}{p^2+1}\right] = \frac{\mathrm{e}^{\mathrm{i}t}}{2\mathrm{i}} + \frac{\mathrm{e}^{-\mathrm{i}t}}{2(-\mathrm{i})} = \sin t.$$

Example 5.2. Solve the initial-value problem

$$x''(t) - 2x'(t) + 2x(t) = 2\mathrm{e}^t \cos t, \qquad x(0) = 0, \quad x'(0) = 0.$$

Solution. Transforming ($\mathcal{L}[x''] = p^2X$, $\mathcal{L}[x'] = pX$, $\mathcal{L}\left[2\mathrm{e}^t\cos t\right] = \dfrac{2(p-1)}{(p-1)^2+1}$) gives

$$X(p) = \frac{2(p-1)}{\left[(p-1)^2+1\right]^2}.$$

Method 1 (residues). The points $p = 1 \pm \mathrm{i}$ are poles of **order two**, so (5.5) applies. For $p_0 = 1 + \mathrm{i}$,

$$\operatorname{Res}\left[X(p)\mathrm{e}^{pt}, 1+\mathrm{i}\right] = \lim_{p\to 1+\mathrm{i}} \frac{\mathrm{d}}{\mathrm{d}p}\left[\frac{2(p-1)\mathrm{e}^{pt}}{(p-1+\mathrm{i})^2}\right],$$

and after differentiating, substituting $p = 1 + \mathrm{i}$ and adding the residue at $p = 1 - \mathrm{i}$, the answer is again $\mathrm{e}^t t\sin t$. The differentiation and substitution for a second-order pole are laborious — that is the drawback of this route.

Method 2 (properties). Put $q = p - 1$; what has to be inverted is $\mathcal{L}^{-1}\left[\frac{2q}{(q^2+1)^2}\right]$. By the derivative-of-the-image property, $\dfrac{\mathrm{d}}{\mathrm{d}p}\dfrac{1}{p^2+1} = -\dfrac{2p}{(p^2+1)^2}$, i.e.

$$\mathcal{L}\left[t\sin t\right] = -\left(\frac{1}{p^2+1}\right)' = \frac{2p}{(p^2+1)^2},$$

so $\mathcal{L}^{-1}\left[\frac{2q}{(q^2+1)^2}\right] = t\sin t$. Then the shift property $\mathcal{L}^{-1}\left[F(p-p_0)\right] = \mathrm{e}^{p_0 t}\mathcal{L}^{-1}\left[F(p)\right]$ with $p_0 = 1$ restores $q \to p - 1$:

$$x(t) = \mathrm{e}^t\, t\sin t.$$

Both methods agree; whenever the properties suffice, the residues are not worth the work.

5.2 Applications of the Laplace Transform

Solving a linear differential (integral) equation by the Laplace transform takes three steps:

1. transform the equation, turning the differential (integral) equation into an **algebraic** equation for the image function;
2. solve that algebraic equation for $X(p)$;
3. invert $X(p)$ to obtain the solution.

Step 3 usually splits $X(p)$ into table entries by partial fractions, or uses the residue formula of §5.1.2 (5.3).

5.2.1 Linear differential (and integral) equations

Example 5.3. Solve the differential-integral equation

$$x'(t) - 2\int_0^t u(s)\,x(t-s)\,\mathrm{d}s + 3\int_0^t x(s)\,\mathrm{d}s = u(t-1), \qquad x(0) = 0,$$

where $u(t)$ is the unit step function.

Solution. Transform the four terms separately:

- first term: $\mathcal{L}\,[x'] = pX(p) - x(0) = pX$;
- second term: $\displaystyle\int_0^t u(s)x(t-s)\,\mathrm{d}s = u(t) * x(t)$ is a convolution, so by the convolution theorem $\mathcal{L}\,[u * x] = \dfrac{X(p)}{p}$ (since $\mathcal{L}[u] = \dfrac{1}{p}$);
- third term: by the integration property $\mathcal{L}\left[\displaystyle\int_0^t x(s)\,\mathrm{d}s\right] = \dfrac{X(p)}{p}$;
- right-hand side: $\mathcal{L}\,[u(t-1)] = \dfrac{\mathrm{e}^{-p}}{p}$.

Hence

$$pX - \frac{2}{p}X + \frac{3}{p}X = \frac{\mathrm{e}^{-p}}{p},$$

and multiplying by p gives $\left(p^2+1\right)X = \mathrm{e}^{-p}$, i.e.

$$X(p) = \frac{\mathrm{e}^{-p}}{p^2+1}.$$

By the delay property the factor e^{-p} simply shifts $\sin t$ to the right by 1:

$$x(t) = u(t-1)\sin(t-1).$$

The factor $u(t-1)$ cannot be omitted: for $t < 1$ both the right-hand side and the initial data vanish, so the solution must vanish too, while for $t \geq 1$ it is a sine oscillation of unit amplitude.

The same three steps cover systems of constant-coefficient equations (reduced to a linear algebraic system for the images), boundary-value problems with a condition at one end only (the unknown initial value at the other end is carried as a parameter and fixed afterwards), and circuit equations such as the series RLC circuit (where the inductance and capacitance contribute LpI and $I/(Cp)$).

5.2.2 Generalized integrals

The integration and differentiation properties of the image function yield two classes of generalized integrals.

Type 1 (containing $1/t$). Let $F(p) = \mathcal{L}[f(t)]$. The integration property of the image function gives $\mathcal{L}\left[\dfrac{f(t)}{t}\right] = \displaystyle\int_p^{+\infty} F(s)\,\mathrm{d}s$. Letting $p \to 0^+$ along the real axis inside the region of convergence, and using the continuity of the Laplace integral at a point of convergence (Abel's theorem),

$$\int_0^{+\infty} \frac{f(t)}{t}\,\mathrm{d}t = \lim_{p\to 0^+} \int_p^{+\infty} F(s)\,\mathrm{d}s = \int_0^{+\infty} F(s)\,\mathrm{d}s.$$

Setting $\mathrm{e}^{-pt} = 1$ in the integrand is not a licence taken lightly: its legitimacy is exactly Abel's theorem.

Example 5.4. Evaluate the Dirichlet integral $\displaystyle\int_0^{+\infty} \frac{\sin t}{t}\,\mathrm{d}t$.

Solution. Take $f(t) = \sin t$, $F(p) = \dfrac{1}{p^2+1}$; by Type 1,

$$\int_0^{+\infty} \frac{\sin t}{t}\,\mathrm{d}t = \int_0^{+\infty} \frac{\mathrm{d}s}{1+s^2} = \arctan s\big|_0^{+\infty} = \frac{\pi}{2}.$$

Type 2 (containing t^n). The derivative-of-the-image property gives

$$F^{(n)}(p) = (-1)^n \int_0^{+\infty} t^n f(t)\,\mathrm{e}^{-pt}\,\mathrm{d}t, \quad \Longrightarrow \quad \int_0^{+\infty} t^n f(t)\,\mathrm{d}t = (-1)^n F^{(n)}(0) \quad (p \to 0),$$

where the step to the right needs F analytic on $\operatorname{Re} p \geq 0$ and the integral convergent.

Example 5.5. Evaluate $\displaystyle\int_0^{+\infty} t^2 \mathrm{e}^{-\alpha t} \sin \beta t\,\mathrm{d}t \quad (\alpha, \beta > 0)$.

Solution. Take $f(t) = \mathrm{e}^{-\alpha t}\sin\beta t$, $F(p) = \dfrac{\beta}{(p+\alpha)^2+\beta^2}$; by Type 2 with $n = 2$,

$$\int_0^{+\infty} t^2 \mathrm{e}^{-\alpha t} \sin \beta t\,\mathrm{d}t = F''(0) = \left.\frac{-2\beta\left[\beta^2 - 3(p+\alpha)^2\right]}{\left[(p+\alpha)^2+\beta^2\right]^3}\right|_{p=0} = \frac{2\beta\left(3\alpha^2-\beta^2\right)}{\left(\alpha^2+\beta^2\right)^3}.$$

For $\beta > \sqrt{3}\,\alpha$ the result is negative: the rapid oscillation of $\sin\beta t$ makes the positive and negative parts almost cancel. This is not a computational slip.

Complex-parameter method. Writing $\sin\beta t = \operatorname{Im} \mathrm{e}^{\mathrm{i}\beta t}$, the integral is the image function evaluated at $p = \alpha - \mathrm{i}\beta$:

$$\int_0^{+\infty} t^2 \mathrm{e}^{-\alpha t}\mathrm{e}^{\mathrm{i}\beta t}\,\mathrm{d}t = \left.\frac{2}{p^3}\right|_{p=\alpha-\mathrm{i}\beta} = \frac{2}{(\alpha-\mathrm{i}\beta)^3},$$

whose imaginary part reproduces the result above. In general, for real-valued f and α inside the region of convergence,

$$\int_0^{+\infty} f(t)\,\mathrm{e}^{-\alpha t}\sin\beta t\,\mathrm{d}t = \operatorname{Im} F(\alpha - \mathrm{i}\beta), \qquad \int_0^{+\infty} f(t)\,\mathrm{e}^{-\alpha t}\cos\beta t\,\mathrm{d}t = \operatorname{Re} F(\alpha - \mathrm{i}\beta).$$

This is the standard way to evaluate generalized integrals carrying a damping factor $\mathrm{e}^{-\alpha t}$: evaluate the image function at a complex parameter, then take its imaginary or real part.

This completes the integral-transform part: the definitions, common pairs and properties of the Fourier and Laplace transforms, and their inversion and applications. The next chapter begins the theory of equations of mathematical physics — the wave, heat and potential equations.